

Seelt

Crop Leaf Disease Classification using Deep Learning

A Project Report

Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Technology in Computer Science and Engineering (AI & Data Engineering)

Submitted by

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Certificate

This is to certify that the project report titled “Seelt – Crop Leaf Disease Classification using Deep Learning and Transfer Learning” has been carried out by Aastha Bhalla, a student of B.Tech (Computer Science and Engineering – AI & Data Engineering) at Lovely Professional University, as part of the requirements for the completion of the said programme.

The work presented in this report is an original piece of work carried out under supervision and has not been submitted elsewhere for the award of any other degree or diploma.

Date:23-07-26

Place: Lovely Professional University, Punjab

Signature of Supervisor

Declaration

I hereby declare that the project titled “Seelt – Crop Leaf Disease Classification” is an original work carried out by me. All the data, code, experiments, and results presented in this report are based on my own effort, using the publicly available PlantVillage dataset, and have not been copied from any other source except where explicitly cited.

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Abstract

Crop diseases are one of the leading causes of yield loss in agriculture, and early, accurate diagnosis is essential to minimise damage and reduce dependence on indiscriminate pesticide use. Manual inspection by agricultural experts is accurate but slow, expensive, and not scalable to the needs of small and marginal farmers. This project, Seelt, presents an end-to-end deep learning system that automatically classifies crop leaf diseases from a single photograph.

The system is built around a MobileNetV2 convolutional neural network, fine-tuned using transfer learning on the PlantVillage dataset, which contains over 54,000 labelled leaf images spanning 38 classes across 14 crop species. The final trained model achieves a validation accuracy of 96.68% and a held-out test accuracy of 97.68%. To make the system more robust to unrelated or accidental uploads, a secondary binary “leaf detector” model is used as a pre-filter that distinguishes genuine leaf images from non-leaf images before the disease classifier is invoked, and a confidence-threshold mechanism withholds a diagnosis when the model is not sufficiently confident.

The complete pipeline is deployed as an interactive Streamlit web application that allows a user to upload a leaf photograph and instantly receive the predicted disease, a confidence score, the top-3 alternative predictions, and detailed reference information including symptoms, treatment, and prevention measures for the diagnosed condition. This report documents the dataset, preprocessing pipeline, model architecture, training methodology, evaluation results, and the design of the deployed application.

Keywords: Plant Disease Classification, Deep Learning, Transfer Learning, MobileNetV2, PlantVillage, Convolutional Neural Network, Streamlit, Precision Agriculture.

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Chapter 1: Introduction

1.1 Background

Agriculture forms the backbone of food security for a large part of the world's population, and crop health directly determines yield and income for farmers. Plant diseases caused by fungi, bacteria, and viruses are responsible for significant annual crop losses worldwide. Traditionally, disease diagnosis has depended on visual inspection by agronomists or experienced farmers, a process that is slow, subjective, and often inaccessible to smallholder farmers in remote areas.

Advances in computer vision and deep learning, particularly Convolutional Neural Networks (CNNs), have made it possible to automate visual diagnosis tasks with high accuracy. Pretrained architectures such as MobileNetV2 offer a lightweight yet powerful backbone that can be fine-tuned on a specific domain, such as plant pathology, using a relatively modest amount of labelled data and computation — making them well suited to real-world, low-resource deployment.

1.2 Problem Statement

Given a photograph of a single crop leaf, the goal of this project is to automatically determine whether the leaf is healthy or affected by a specific disease, and to communicate this diagnosis — along with symptoms, causes, treatment, and prevention advice — to a non-expert user in an accessible, easy-to-use interface. The system must also be able to reject or flag inputs that are not valid leaf images, to avoid producing misleading diagnoses on irrelevant photos.

1.3 Objectives

- To study and prepare the PlantVillage dataset for supervised image classification.
- To design an efficient image preprocessing and augmentation pipeline suited to MobileNetV2.
- To fine-tune a MobileNetV2 model using transfer learning to classify 38 crop-disease categories.
- To build an auxiliary leaf-vs-non-leaf detector that improves the robustness of the overall system.
- To evaluate the trained model using standard classification metrics on an independent test split.
- To deploy the complete pipeline as an interactive, user-friendly Streamlit web application.

1.4 Scope of the Project

The current scope of the project covers image-based classification of 38 disease/healthy categories across 14 crop species using the “color” variant of the PlantVillage dataset. The grayscale and segmented variants of the dataset are also verified and explored for future comparative work but are not used in the final deployed model. The application is designed for single-image, single-leaf inference and is intended as a decision-support tool rather than a replacement for expert agronomic diagnosis.

Chapter 2: Literature Review

Automated plant disease detection using image processing and machine learning has been an active area of research for over a decade. Early approaches relied on hand-crafted features — colour histograms, texture descriptors such as GLCM, and shape features — combined with classical classifiers like Support Vector Machines (SVMs) and Random Forests. While these approaches demonstrated feasibility, they were sensitive to lighting conditions, background clutter, and required extensive manual feature engineering.

The introduction of deep Convolutional Neural Networks (CNNs) shifted the field towards end-to-end learned representations. Architectures such as AlexNet, VGG, ResNet, and Inception achieved strong results on large-scale image classification benchmarks and were subsequently adapted to plant disease datasets, most notably the PlantVillage dataset, which has become a standard benchmark for this task due to its scale and class diversity.

Transfer learning — reusing weights learned on a large generic dataset such as ImageNet, and fine-tuning only the final layers on a smaller domain-specific dataset — has been widely shown to reduce training time and improve generalisation, especially where labelled agricultural data is comparatively limited. Lightweight architectures such as MobileNetV1/V2 were specifically designed to trade a small amount of accuracy for a large reduction in parameter count and computational cost, making them attractive for deployment on mobile devices or low-power edge hardware used in field conditions.

This project builds directly on this line of work, adopting MobileNetV2 for its favourable accuracy-to-efficiency trade-off, and extends the standard single-model pipeline with an additional leaf/non-leaf gating classifier and a confidence-threshold rejection mechanism — both aimed at making the system more reliable when used by real, non-expert users outside a controlled dataset environment.

Chapter 3: Dataset Description

3.1 The PlantVillage Dataset

The project uses the publicly available PlantVillage dataset, one of the most widely used benchmark datasets in the plant disease classification literature. The dataset contains close-up photographs of individual crop leaves captured under laboratory conditions, organised into folders where each folder name encodes the crop and disease label in the form “Crop__Disease” (for example, Apple__Apple_scab or Tomato__healthy).

The dataset spans 14 crop species — Apple, Blueberry, Cherry, Corn (Maize), Grape, Orange, Peach, Bell Pepper, Potato, Raspberry, Soybean, Squash, Strawberry, and Tomato — and 38 total classes, since several crops have multiple disease categories in addition to a healthy class. Each image is originally 256 × 256 pixels in JPEG format.

3.2 Dataset Variants

The raw PlantVillage release is provided in three parallel variants, all of which were verified as part of the initial data-check stage of this project:

- Color — the original RGB leaf photographs.
- Grayscale — the same photographs converted to single-channel grayscale.
- Segmented — the leaf isolated from its background using image segmentation.

Table 3.1: Dataset variants and approximate image counts

Dataset Variant	Classes	Approx. Images
Color	38	54,295
Grayscale	38	54,303
Segmented	38	54,306

All three variants contain the same 38 classes and a comparable number of images. This project's model is trained and evaluated on the Color variant; the grayscale and segmented variants were verified for completeness and are earmarked for future comparative experiments (see Chapter 8).

3.3 Class Distribution

The dataset is moderately imbalanced across classes — for example, Orange__Haunglongbing_(Citrus_greening) contains 5,507 images while Potato__healthy contains only 152. A representative sample of class-wise counts from the Color dataset is shown below.

Table 3.2: Sample class-wise image distribution (color dataset)

Class	Images
Apple___Apple_scab	622
Apple___Black_rot	621
Apple___Cedar_apple_rust	275
Apple___healthy	1,645
Blueberry___healthy	1,502
Corn_(maize)___Common_rust_	1,192
Grape___Esca_(Black_Measles)	1,383
Orange___Haunglongbing_(Citrus_greening)	5,507
Potato___healthy	152
Soybean___healthy	5,090
Tomato___Bacterial_spot	2,127

This class imbalance is a known characteristic of the PlantVillage dataset and is an important consideration when interpreting per-class accuracy; classes with very few samples are more prone to noisy evaluation metrics.

Chapter 4: System Design and Methodology

4.1 Overall Workflow

The project follows a standard supervised deep learning pipeline, extended with a secondary gating model for deployment robustness. The end-to-end workflow proceeds as follows: dataset verification, exploratory data analysis, image preprocessing and augmentation, dataset/loader construction, transfer-learning-based model training, model evaluation on a held-out test split, and finally deployment through a Streamlit web application in which a leaf/non-leaf detector gates the disease classifier.

4.2 Data Preprocessing

Every image is resized to 224×224 pixels — the input resolution expected by MobileNetV2 — converted to a PyTorch tensor, and normalised using the standard ImageNet channel-wise mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. This normalisation is essential because MobileNetV2's convolutional backbone was pretrained on ImageNet and expects inputs in this statistical range. The identical preprocessing pipeline is reused at inference time in `predict.py` and `leaf_detector.py` to guarantee train/inference consistency.

4.3 Data Augmentation

To improve generalisation and reduce overfitting given the moderate size of some classes, random augmentations are applied to the training split only; validation and test images are resized and normalised without any random transformation, so that evaluation remains fair and reproducible.

Table 4.1: Data augmentation operations applied during training

Operation	Parameter	Purpose
Resize	224×224 px	Match MobileNetV2's expected input size
Random Horizontal Flip	$p = 0.5$	Simulate leaf orientation variability
Random Rotation	$\pm 15\text{--}20^\circ$	Simulate camera-angle variation
ToTensor	—	Convert PIL image to a normalised PyTorch tensor
Normalize	ImageNet mean/std	Match the statistics MobileNetV2 was pretrained on

4.4 Transfer Learning with MobileNetV2

MobileNetV2 is a lightweight CNN architecture built around depthwise-separable convolutions and inverted residual blocks with linear bottlenecks, designed to deliver strong accuracy at a fraction of the

parameter count and computational cost of larger architectures such as VGG or ResNet. This makes it particularly well suited to a project that may ultimately be deployed on modest hardware.

In this project, MobileNetV2 is initialised with ImageNet-pretrained weights. The original 1000-class classification head is replaced with a new fully-connected layer mapping the 1,280-dimensional feature vector to 38 output classes. During training, the convolutional feature-extractor layers are frozen (their weights are not updated), and only the newly added classifier layer is trained. This is a standard “feature extraction” style of transfer learning, which is fast to train and effective when the target domain (leaf images) shares low-level visual structure with the pretraining domain (natural images).

4.5 Leaf Detector Module

Because the disease classifier is only trained to distinguish between 38 known crop-disease categories, it will still confidently assign one of these labels to an image of anything else — a hand, a face, an unrelated object — if given no filtering. To mitigate this, a second, independently trained binary classifier (“leaf” vs “non_leaf”), also based on a MobileNetV2 backbone with its classification head replaced by a 2-class layer, is used as a pre-filter. Any uploaded image is first passed through this leaf detector; only images classified as “leaf” are forwarded to the disease classifier, while “non_leaf” images are rejected with a clear message to the user.

The leaf-detector's own training data is organised into train/validation/test splits (70% / 15% / 15%) using a dedicated dataset-splitting utility, mirroring the standard split ratio used for the final held-out evaluation of the disease classifier.

Chapter 5: Implementation

5.1 Technology Stack

Table 5.1: Technology stack used in the project

Category	Technology
Programming Language	Python 3.13
Deep Learning Framework	PyTorch
Computer Vision / Pretrained Models	TorchVision (MobileNetV2)
Image Processing	Pillow (PIL)
Data Handling	torch.utils.data (Dataset, DataLoader, random_split)
Visualization / EDA	Matplotlib
Development Environment	Jupyter Notebook
Web Application / Deployment	Streamlit
Version Control	Git & GitHub

5.2 Project / Repository Structure

The codebase is organised into clearly separated notebooks (used for experimentation, training, and evaluation), a src package (containing the reusable, production-facing inference code), and an app package (containing the Streamlit front-end). This separation keeps exploratory notebook work independent from the deployable application code.

Table 5.2: Repository / module structure

File / Module	Responsibility
notebooks/01_data_check.ipynb	Verifies dataset folders, counts images and classes, visual sanity checks
notebooks/02_data_preprocessing.ipynb	Builds and visualises resize/augment/normalize pipeline and DataLoader
notebooks/03_mobilenetv2_training.ipynb	Loads MobileNetV2, freezes backbone, trains classifier head, saves .pth weights
notebooks/04_prediction.ipynb	Loads the trained model and runs a single-image sanity-check prediction
notebooks/05_leaf_detector.ipynb	Trains the binary leaf / non-leaf gating classifier
notebooks/06_test_accuracy.ipynb	Evaluates the final saved model on an independent held-out test split

File / Module	Responsibility
src/preprocess.py	Shared image transform pipeline and class-name lookup used at inference time
src/predict.py	Loads mobilenet_plant_disease_new.pth and returns top-3 disease predictions with confidence
src/leaf_detector.py	Loads leaf_detector.pth and classifies an image as leaf / non_leaf
src/disease_info.py	Static reference data (description, symptoms, treatment, prevention) and confidence-based decision logic
src/split_dataset.py	Splits the leaf-detector dataset into train / val / test folders (70/15/15)
app/app.py	Streamlit web application — Predict, Disease Library, and About tabs
models/*.pth	Saved model weights (disease classifier and leaf detector)

5.3 Model Training

Training is carried out in notebooks/03_mobilenetv2_training.ipynb. The Color dataset is loaded with the augmenting train_transform pipeline, and split into training and validation subsets using an 80/20 random split (43,437 training images and 10,860 validation images). MobileNetV2 is loaded with ImageNet-pretrained weights, its final classifier layer is replaced with a Linear(1280 → 38) layer, and the convolutional feature-extractor parameters are frozen so that only the new classifier head is updated during training.

The model is trained using the Adam optimizer (learning rate = 0.001) with Cross-Entropy loss, over 10 epochs and a batch size of 32, on CPU. Training accuracy improved steadily from 86.00% in epoch 1 to 95.44% by epoch 10, with training loss falling from 0.6443 to 0.1350 over the same period (full epoch-wise figures are given in Chapter 6). The final model weights are saved as mobilenet_plant_disease_new.pth.

5.4 Prediction Pipeline

At inference time, src/predict.py loads the saved model weights, reconstructs the identical MobileNetV2 + Linear(1280, 38) architecture, and applies the same resize / tensor / normalise preprocessing used during training (via src/preprocess.py) to guarantee consistency between training and inference. Given an image, the model produces a softmax probability distribution over the 38 classes, from which the top-3 predictions (class name and confidence percentage) are returned.

src/leaf_detector.py mirrors this same pattern for the binary leaf/non-leaf model, returning the predicted class ("leaf" or "non_leaf") along with a confidence score.

src/disease_info.py then combines the raw prediction with a static, hand-curated knowledge base covering all 38 classes — crop name, human-readable disease name, description, symptoms, treatment, and prevention guidance — and applies a configurable confidence threshold (currently 60%) to decide whether the prediction is reliable enough to present as a full diagnosis, or should instead be shown as a low-confidence result along with the top candidate guesses and image-quality suggestions.

5.5 Web Application (Streamlit)

The trained pipeline is exposed to end users through app/app.py, a Streamlit application named “Seelt” with a custom dark, agriculture-themed visual style. The application is organised into three tabs:

- Predict — the user uploads a JPG/PNG image of a single leaf; the image is first passed through the leaf detector (rejecting non-leaf uploads), then through the disease classifier; results are displayed with a confidence meter, the top-3 predictions, and tabbed sections for Description, Symptoms, Treatment, and Prevention.
- Disease Library — a searchable, filterable reference browser covering all 38 supported classes, allowing a user to look up any disease by crop or keyword even without uploading an image.
- About — project information, model details, headline accuracy figures, and the list of supported crops.

A sidebar additionally surfaces model metadata (architecture, validation accuracy, number of classes, and the active confidence threshold) and the full list of supported crops, giving the user context on the system's capabilities and limitations before they upload an image.

Chapter 6: Results and Evaluation

6.1 Training Results

The classifier head was trained for 10 epochs on the 80% training split (43,437 images) of the Color dataset. Training accuracy rose quickly in the first few epochs and continued to improve gradually thereafter, indicating that the frozen, pretrained MobileNetV2 backbone provided a strong, transferable feature representation for this domain.

Table 6.1: Epoch-wise training loss and accuracy

Epoch	Training Loss	Training Accuracy
1	0.6443	86.00%
2	0.2457	93.47%
3	0.1988	94.24%
4	0.1719	94.74%
5	0.1632	94.91%
6	0.1516	95.21%
7	0.1435	95.43%
8	0.1388	95.40%
9	0.1351	95.52%
10	0.1350	95.44%

6.2 Validation Results

On the 20% validation split (10,860 images) held out from the same random split, the trained model achieved a validation loss of 0.0992 and a validation accuracy of 96.68%, indicating good generalisation with no evidence of severe overfitting given that training and validation accuracy remained close.

6.3 Independent Test Set Evaluation

To obtain an evaluation figure on data organised independently of the training run, notebooks/06_test_accuracy.ipynb reloads the saved mobilenet_plant_disease_new.pth weights and re-splits the full Color dataset using a 70% / 15% / 15% train/validation/test ratio, then evaluates the model on the resulting 8,148-image test subset. This evaluation reports a test loss of 0.0769 and a test accuracy of 97.68%, corroborating the validation results and confirming that the trained model generalises well to unseen images from the same dataset distribution.

Table 6.2: Summary of model evaluation metrics

Evaluation Stage	Split Size	Loss	Accuracy
Training (final epoch)	43,437 images	0.1350	95.44%
Validation (80/20 split)	10,860 images	0.0992	96.68%
Independent Test (70/15/15 split)	8,148 images	0.0769	97.68%

Note: the Streamlit application's "About" tab and sidebar currently display a validation accuracy figure of 94.40%, which reflects an earlier training run/checkpoint of the same architecture; the figures in Table 6.2 are taken directly from the training (03) and test-accuracy (06) notebooks associated with the currently saved mobilenet_plant_disease_new.pth weights, and it is recommended that the displayed figure in the application be refreshed to match the latest evaluation before final submission or deployment.

6.4 Discussion

The consistently high accuracy across training, validation, and an independently constructed test split suggests the model has learned discriminative, generalisable visual features for the 38 target classes rather than memorising the training set. As is common with the PlantVillage dataset, some of this strong performance can be attributed to the relatively controlled, consistent image-capture conditions of the dataset (uniform backgrounds, consistent lighting); real-world field photographs with cluttered backgrounds, variable lighting, and partial occlusion are typically more challenging, which is precisely the gap the leaf-detector pre-filter and confidence-threshold mechanism are designed to partially address in the deployed application.

Chapter 7: Conclusion and Future Scope

7.1 Conclusion

This project successfully demonstrates an end-to-end deep learning pipeline for crop leaf disease classification, from raw dataset verification through to a deployed, interactive web application. By fine-tuning a MobileNetV2 model via transfer learning on the PlantVillage dataset, the system achieves a test accuracy of 97.68% across 38 disease/healthy classes spanning 14 crop species. The addition of a dedicated leaf/non-leaf detector and a confidence-based rejection mechanism makes the deployed system meaningfully more robust to the kind of unpredictable, unfiltered input a real end user is likely to provide, compared to a bare single-model classifier.

The resulting Streamlit application, Seelt, packages this pipeline into an accessible tool that not only returns a diagnosis, but also educates the user with symptoms, treatment, and prevention guidance for the identified condition — directly supporting the project's original goal of assisting farmers and researchers in early disease detection.

7.2 Limitations

- The model is trained and validated entirely on the PlantVillage dataset, which was captured under relatively controlled conditions; performance on real field photographs with cluttered backgrounds, poor lighting, or multiple overlapping leaves has not been formally evaluated.
- Some classes (e.g. Potato___healthy with only 152 images) are significantly under-represented compared to others (e.g. Orange___Haunglongbing with 5,507 images), which may bias per-class performance.
- The confidence threshold used for accept/reject decisions (60%) is a manually chosen heuristic rather than one tuned against a labelled “out-of-distribution” validation set.

7.3 Future Scope

- Extend training and evaluation to the grayscale and segmented dataset variants already verified in this project, and compare their performance against the color-image model.
- Fine-tune deeper layers of the MobileNetV2 backbone (rather than only the classifier head) or experiment with larger architectures to assess potential accuracy gains.
- Collect or incorporate a set of real-world field images to evaluate and improve robustness beyond the PlantVillage domain.
- Add Grad-CAM or similar visual explanation overlays so users can see which regions of the leaf most influenced a given prediction, improving trust and interpretability.
- Package the model for on-device / mobile inference, leveraging MobileNetV2's lightweight design for offline use in the field.
- Formally evaluate precision, recall, and F1-score on a per-class basis, and use a confusion matrix to identify commonly confused disease pairs.

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Appendix

Appendix A: Supported Crops and Disease Classes

The model recognises 38 total classes across the following 14 crop species (Table 8.1). A full class list, including exact PlantVillage folder/label names, is available in `src/disease_info.py`.

Table 8.1: Supported crops and number of associated classes

Crop	Number of Classes (incl. healthy)
Apple	4
Blueberry	1
Cherry (including sour)	2
Corn (maize)	4
Grape	4
Orange	1
Peach	2
Bell Pepper	2
Potato	3
Raspberry	1
Soybean	1
Squash	1
Strawberry	2
Tomato	10

Appendix B: Model Configuration Summary

- Base architecture: MobileNetV2 (ImageNet-pretrained)
- Modified layer: `classifier[1] → nn.Linear(1280, 38)` for disease model / `nn.Linear(1280, 2)` for leaf detector
- Frozen layers: all convolutional feature-extractor layers (`features.parameters()`)
- Trainable layers: final classifier layer only
- Loss function: CrossEntropyLoss
- Optimizer: Adam, learning rate = 0.001
- Batch size: 32
- Input resolution: $224 \times 224 \times 3$
- Epochs: 10 (disease classifier)

- Saved weights: models/mobilenet_plant_disease_new.pth, models/leaf_detector.pth

Github_link: <https://github.com/aash0617/Seelt-Crop-Disease-Classification>

Video link: <https://www.loom.com/share/0c2b39ecc393495797b33a301e2bcdd9>



